UX Evaluation & Redesign

Improving Digital User Flows

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UX Evaluation

1. Product Selection (Mobile or Web App)

The Application selected for reviewing and finding issues is "**Gamyam**" a Mobile App, which is by TGSRTC – a Live Bus Tracking Application.

The TGSRTC Gamyam app provides real-time tracking for Telangana State Road Transport Corporation buses. Commuters can view live bus locations, estimated arrival times (ETAs), and nearby bus stops. It tracks Pushpak AC Airport coaches, district specials, and city metro services. Users highly praise the utility of the app for cutting down wait times.

Core Features

- **Live Route Tracking:** Search by bus number or route to see live map tracking and active trips.
- **Safety Alerts:** Includes a "Flag a Bus" feature that allows women traveling after 6:30 PM to send a pickup alert to drivers, and an SOS button to contact emergency services.
- **Multilingual Support:** Fully available in both English and Telugu.

Platforms and Download

The Gamyam app is available for free across major mobile devices:

- Download on the Google Play Store.
- Download on the Apple App Store.

2. User Flow Selection

Two user flows have been selected for reviewing. The aim of user flows is searching for bus availability.

i. Airport Services
   a. Search by Route – From location and To location
   b. Search by Route Number
   c. Search by Bus Number

ii. City Bus Services
   a. Search by Route – From location and To location
   b. Search by Route Number
   c. Search by Bus Number

3. Screenshots of Current Experience

Case 1: City Bus Services

The below figures represent the app UI displaying various options available for performing the necessary actions by users.

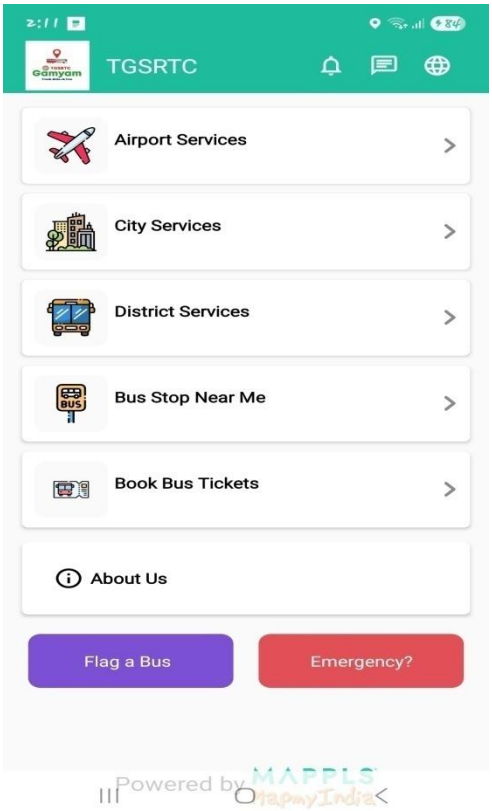


Fig 3.1 Homepage consisting of various services of buses

Here, one issue identified is up on tapping the Notification Icon – Notifies nothing.

Impact:

Creates an ambiguity and confusion for user on identifying what exactly this option works for.

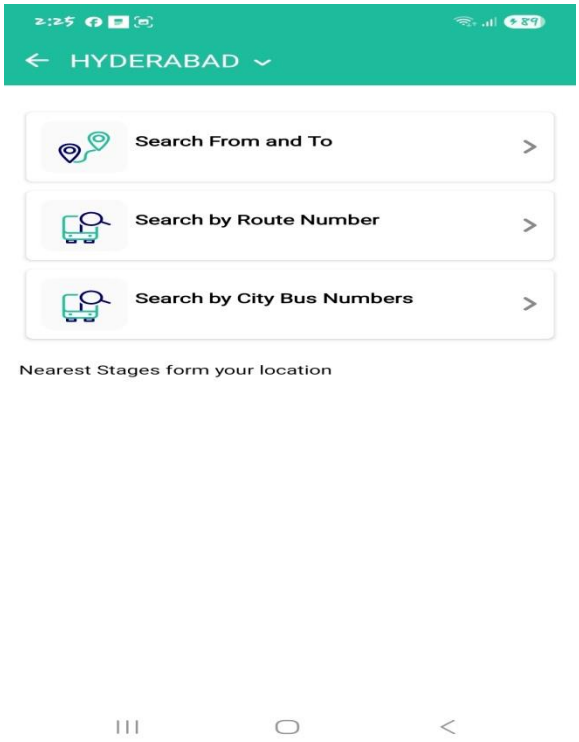


Fig 3.2 Options for finding Bus

The user flow gets initiated by choosing any of these options available according to the knowledge of buses the user has.

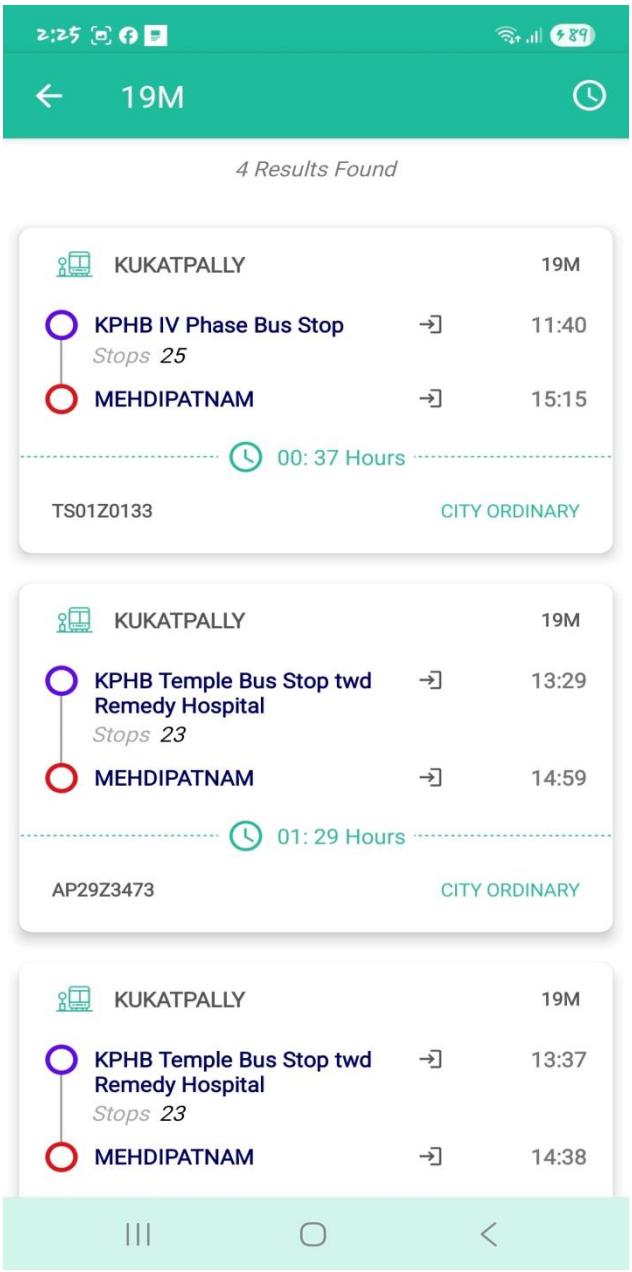


Fig 3.3 Search by Route Number – Here, 19M is the Route Bus Number

Scenario 1:

When the search of the bus is done using the option – ***'Search by Route Number'*** (Here Route Number = 19M, that serves between KPHB and Mehdipatnam and the travel time in reality is 2Hrs ~approx), one bus shown improper details.

According to the interface designed, the app has to display only the buses that are currently running in the specified route. But in the above figure, considering the first bus displayed, there

is an ambiguity in its drop-off timing and duration, though the status is running it is actually not running.

Impact:

This creates lack of clarity about the buses shown as active and running.

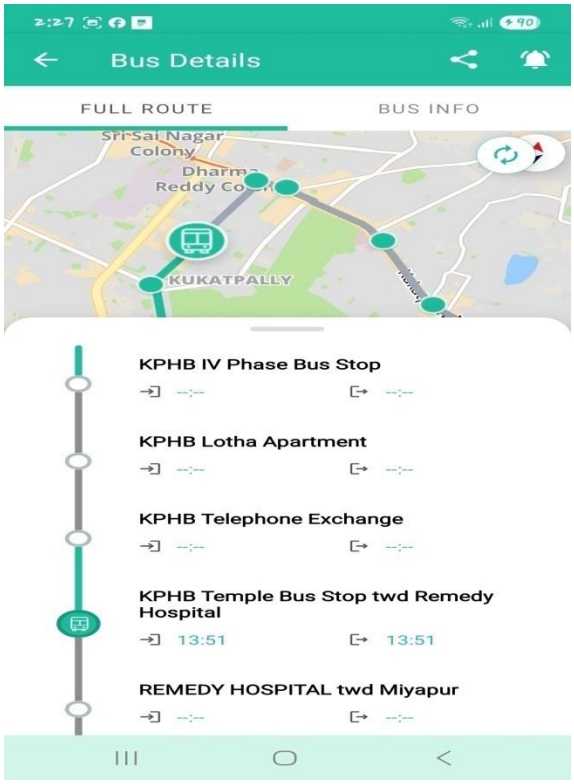


Fig 3.4 Indication of Wrong Time, No previous stops' bus status – "Bus Left"

According to the figure 3.3, as the app has to display only the buses that are currently running in the specified route, but when selected to view and track the bus, it neither displayed right time nor right location and also no previous stop status of arrival and departure times.

Impact:

This again creates lack of clarity about the buses shown as active and running.

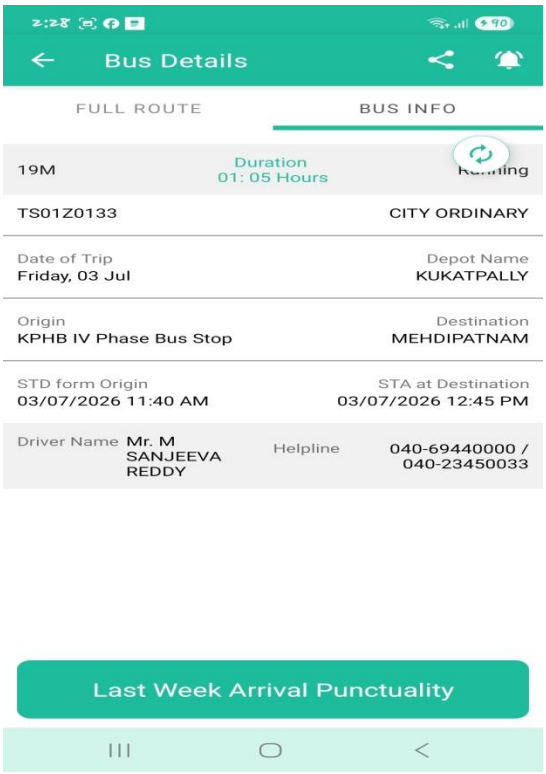


Fig 3.5 Bus Info – showing current status as "Running"

In the above figure, though the status is shown as Running, the bus is not actually moving since the time displayed of Origin and Destination are mismatch and location shown are wrong as per Fig 3.4

Impact:

This again creates lack of clarity about the buses shown as active and running.

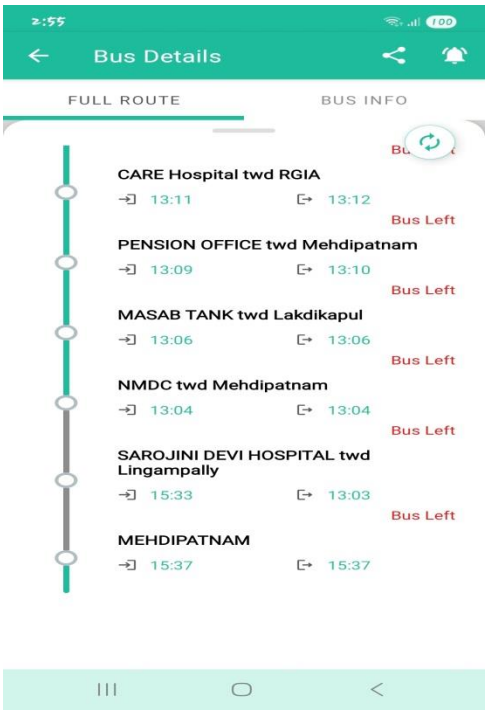


Fig 3.6 Wrong Time & Bus Status

The same bus when tracked for various stops in the route indicates the status as Bus Left but with irrelevant times again which is shown in above figure and the figure illustrates - to identify and confirm that bus is not running, the time displayed, the location or stops left shall remain unclear which again acts as a proof.

Impact:

This creates lack of clarity about the buses shown as active and running.

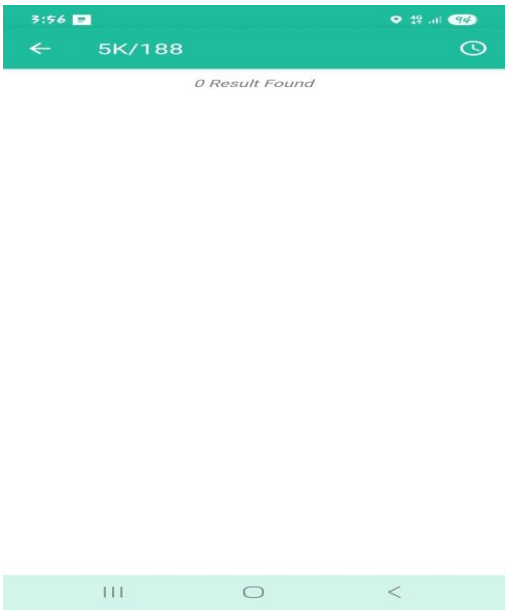


Fig 3.8 Results Not Found Case

When served with different bus number for a route (here, 5K/188 which should display buses from Mehdipatnam and Shamshabad) the interface displayed '0 Results Found'

Impact:

This will leave user having no idea on deciding whether to wait for another bus or choose any other alternatives to reach.

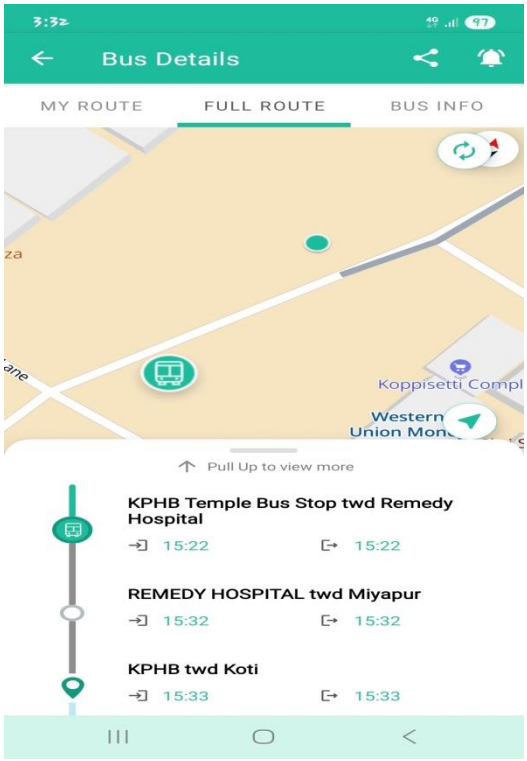


Fig 3.7 Live Location Mismatch

Scenario 2:

When the search of the bus is done using the option – ***'Search by Route'***, one bus shown improper location details. The above figure conveys the actual bus location and wrong live status indicated.

Impact:

This again creates lack of clarity about the buses shown as active and running.

Scenario 3:

When the search of the bus is done using the option – ***'Search by City Bus Number'***, No User is aware of the vehicle numbers to search for buses.

Impact:

Option failed in finding buses.

Case 2: Airport Services

The Search of buses to RGIA (Rajiv Gandhi International Airport, Shamshabad) from various major routes is performed with three different options of search available. The below figure depicts the search made by using 'Search From and To', providing the boarding point location and dropping point. But here, since the homepage specified the Airport Services, which by default meant that buses reach Airport from various locations and vice versa and hence this option need to be redesigned.

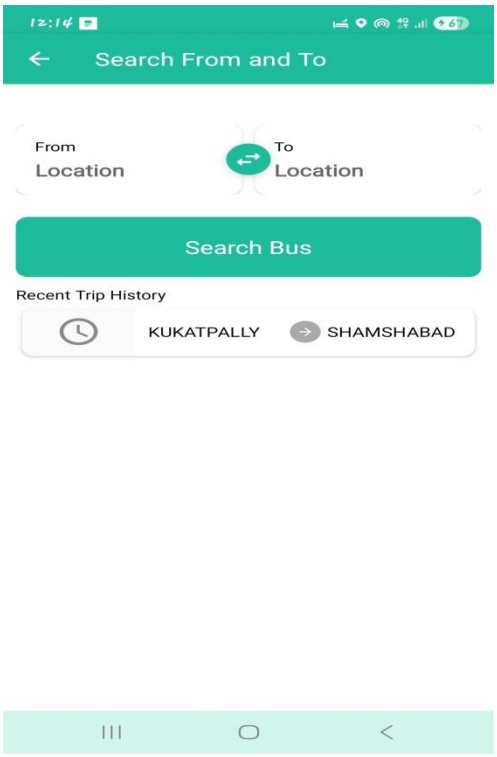


Fig 3.8 Inconsistency in location when already selected Airport Service (default)

Impact:

Creates ambiguity making user to again enter the 'To' location when user wanted to reach Airport and enter 'From' when user boarding at Airport.

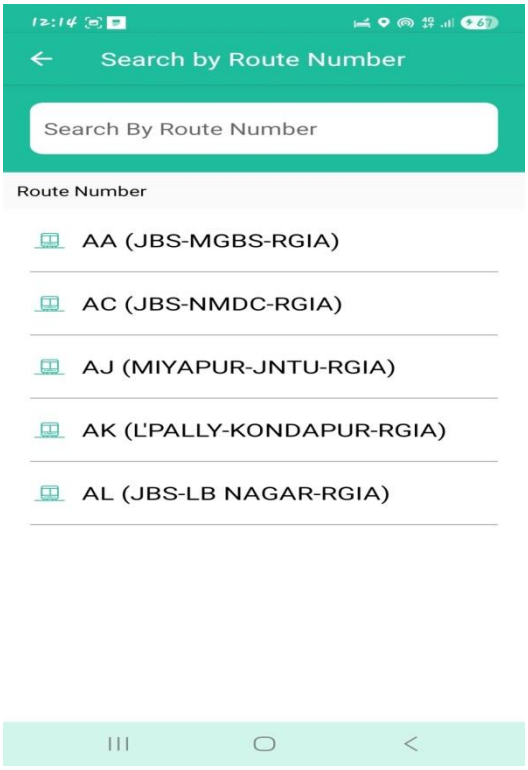


Fig 3.9 Option Name indicated as Route Number (to be in number format)

Impact:

When the interface is asking to enter the Route Number, user's mind would definitely go with entering the bus number if remembers. But in this case, it is asking to enter unique bus identification numbers that are displayed on the screen not even letting the user to choose the below displayed options.

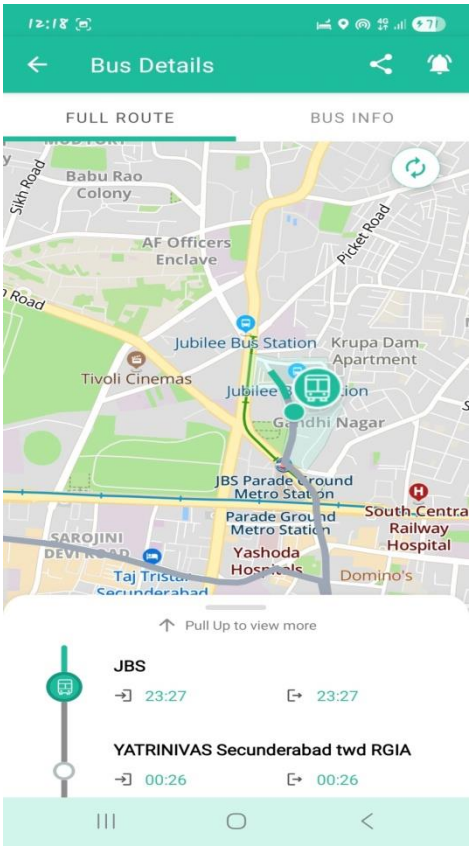


Fig 3.10 Improper Location and Live Status

Impact:

To any passenger who wanted to reach airport, definitely look for live status of the bus and transparency of the location so they can plan their travel time to reach airport. Showing the incorrect location updates make the passenger run out of time in reaching the destination.

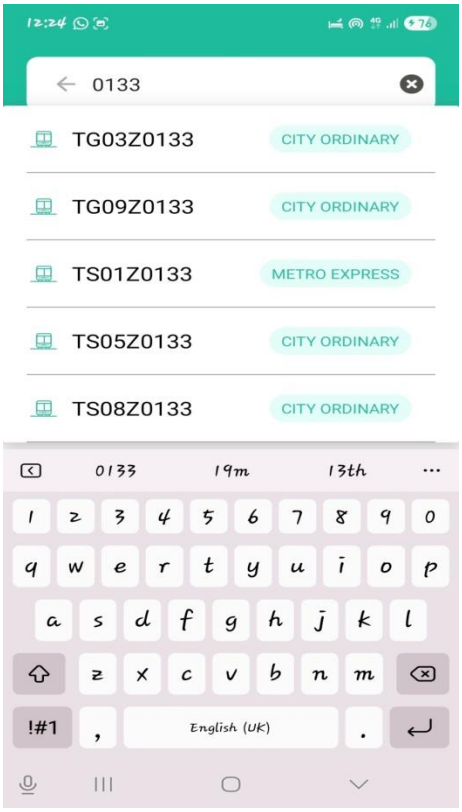


Fig 3.11 Lack of knowledge on Bus Numbers

Impact:

Since Bus Numbers are not known by users, can't be identified and remembered easily. This again leaving the passenger provided with no idea of alternatives. Moreover, the airport services offered are by name called as - Pushpak Buses and not any ordinary, metro express buses which are as shown in the figure.

4. List of Usability Issues (Minimum 5)

To evaluate the usability of Gamyam, TGSRTC's live bus tracking application, the bus search user flows under both City Bus Services and Airport Services were reviewed in detail. This involved walking through the app as a typical commuter would — searching for buses by route, route number, and bus number — and closely observing the interface at each step of the journey. Through this hands-on review, six key usability issues were identified, ranging from unresponsive notifications and input validation gaps to unreliable live tracking data, unhelpful search outcomes, and the absence of any help or guidance for users. These issues are detailed below.

Issue #1: Notification icon failure, as it notified nothing upon tapping.

Issue #2: Option Name and Input mismatch – Asked to input number but text format is accepted

Issue #3: Lack of knowledge for the user in using option called Search by City Bus Number – to give vehicle number as input which is not identified by the user anytime

Issue #4: Live Tracking error, Travel Duration displayed error, drop-off time displayed error

Issue #5: 0 results found, rather than displaying relevant information

Issue #6: Help & Documentation not provided.

5. Perform UX Evaluation

A. Usability Heuristics

- Identified **5 usability issues** using Nielsen's principles

Out of 10 Usability Heuristics, the issues that are found failed in satisfying the below:

1st Heuristic – Visibility of System Status

Failed in:

- Notifications functionality – Notifying nothing
- Airport Services upon selection made initially, yet asking for 'To Location'

4th Heuristic – Consistency and Standards

- Under Airport Services,
  - Route Number is given in Textual format rather than Number.
  - Bus Number (Vehicle Number) is also not known to the user.
- Under City Bus Services,
  - Bus running status displayed is active, but bus is not schedule - Route Error.
  - Bus Numbers (Vehicle Number) are not known to users which are indeed not identified easily.

5th Heuristic – Error Prevention

- Under City Bus Services,
  - Live Tracking, Travel Duration displayed error, drop-off time displayed error

9th Heuristic – Recognize, Diagnose and Recovery

- Search resulted with – '0 results found.'

10th Heuristic – Help & Documentation

- FAQs, Help Button not provided.

B. UX Laws - Mapping at least **3 issues** to the UX laws:

Out of the 15 laws, 3 distinct laws are covering these 6 issues:

- **Postel's Law**
- **Jakob's Law**
- **Information Scent**

Issue 1: Notification Icon Failure

Problem

The notification icon appears interactive, but tapping it produces no response, feedback, or message.

UX Laws -

1. Jakob's Law

Users expect notification icons to behave consistently with other mobile applications.

Expected behaviour:

- Badge count
- Notification list
- "No notifications" message

Instead, nothing happens.

2. Information Scent

The notification icon signals that information is available. Since tapping it provides no feedback, the information cue is misleading.

Issue 2: Option Name and Input Mismatch

Problem

The field says (under Airport Services),

Search by City Bus Number

But accepts text format

AC (JBS-NMDC-RGIA)

AA (JBS-MGBS-RGIA)

AJ (Miyapur-JNTU-RGIA)

AK(L’PALLY-Kondapur-RGIA)

AL (JBS-LB Nagar-RGIA)

although only numbers should be entered according to the option name.

Note: The abbreviated forms are the main bus route starting and ending via stop. JBS – Jubliee Bus Stand, MGBS – Mahatma Gandhi Bus Station, RGIA – Rajiv Gandhi International Airport, L'Pally – Lingam pally, NMDC – National Mineral Development Corporation)

UX Law - Postel's Law

Systems should accept reasonable input but validate it intelligently.

If only numbers are valid, the system should:

- Display numeric keyboard
- Reject letters immediately
- Explain the expected format

Issue 3: "Search by City Bus Number" is Confusing

Problem

Most passengers know

- Route Number
- Boarding Stop
- Destination

They rarely know the **vehicle registration number**.

This forces users to recall information they don't possess.

UX Laws -

Jakob's Law

Transport apps usually allow searching by:

- Route Number
- Source
- Destination

Users naturally expect these options.

Issue 4: Live Tracking, Travel Duration & Drop-off Time Errors

Problem

The application displays:

- Incorrect live location
- Incorrect ETA
- Incorrect travel duration

Users lose confidence because they cannot determine whether the bus is actually delayed or the application is malfunctioning.

UX Laws -

1. Jakob's Law

Users expect real-time tracking similar to Google Maps or Uber/Rapido/Ola, where locations and ETAs update reliably.

2. Information Scent

Current information does not accurately reflect the bus's status, making it difficult for users to trust or interpret the data.

Issue 5: "0 Results Found"

Problem

The application only displays

0 Results Found, without explaining why or suggesting the next step.

UX Laws

1. Postel's Law

A robust system should handle unexpected input gracefully instead of ending the user's journey.

2. Information Scent

Users receive no clues about what caused the issue or how to proceed.

3. Goal-Gradient Effect

The user's progress stops completely, reducing motivation to complete the task.

Issue 6: Help & Documentation Missing

Problem

First-time users receive no guidance on:

- Searching buses
- Difference between Route Number and Vehicle Number
- Live Tracking
- ETA
- Bus status meanings

UX Laws

1. Picture Superiority Effect

Visual illustrations, icons, and examples are easier to understand than lengthy text, especially in a transport app used on the go.

2. Information Scent

Contextual hints should guide users exactly where they need help, reducing guesswork.

3. Jakob's Law

Users expect common support patterns such as FAQs, tooltips, and help icons because these are standard in modern mobile apps.

C. Mapping to UX Heuristics & Laws

Issue	Which Heuristic Violated	How to Improve
1. Notification icon failure (tap does nothing)	Visibility of System Status	Instead of doing nothing: If notifications exist, • Route updated If no notifications, • **No notifications available.** • We'll notify you about bus updates here. This reassures the user that the feature is working.
2. Option Name and Input mismatch (expects number but accepts alphanumeric)	Error Prevention	Option Name to be changed. Ex: Choose the route buses below.
3. Search by City Bus Number is confusing because users	Match Between System & Real World, Recognition	Search by City Bus Number can be ignored.

don't know the vehicle number	Rather than Recall	
4. Live Tracking, Travel Duration & Drop-off Time errors	Visibility of System Status, Recognize/Diagnose/Recover from Errors	Display: • Last Updated: **10:45 AM** • Last Updated 2min ago If tracking is unavailable: ***Live location is temporarily unavailable.***
5. "0 Results Found" instead of meaningful guidance	Recognize, Diagnose & Recover from Errors	Display - • ***No buses are currently available for this route.*** Possible reasons • Service has ended • Route unavailable today • Bus number incorrect You may, • Modify your search This keeps the user moving toward their goal instead of reaching a dead end.
6. Help & Documentation missing	Help & Documentation	Include • How to search buses • Route Number vs Vehicle Number • Live Tracking explained • GPS permission help • Frequently Asked Questions Add ***i*** icons beside unfamiliar fields. Example: Search by Route Number ***i***

6. Improvement Recommendations

6.1 Redesign the Search Section

The current search option, **"Search by City Bus Number,"** is unclear because most users are unfamiliar with the city's vehicle number. Passengers generally know the bus route number or the source and destination rather than the actual vehicle registration number. Instead making the other two options alone to be available would make the feature much easier to understand. Additionally, implementing auto-suggestions while typing and validating the input format would help users find the correct bus quickly while reducing incorrect searches and improving the overall search experience.

6.2 Improve the Notification Experience

The notification icon currently does not provide any response when tapped, leaving users unsure whether the feature is working or if there are simply no notifications available. To improve usability, tapping the notification icon should open a notification panel. If there are no updates, the application should display a friendly message such as **"No new notifications available. We'll notify you about bus delays, route changes, and service updates here."** This reassures users that the feature is functioning correctly and builds confidence in the application.

6.3 Improve the Live Tracking Screen

The current live tracking screen does not clearly indicate whether the displayed information is up to date. Incorrect or outdated travel duration, drop-off time, and bus location can reduce user trust in the application. To improve this experience, the application should display the **last updated time**, provide accurate estimated arrival times, and clearly indicate when live tracking data cannot be retrieved. In cases where tracking fails, the app should explain the issue and offer options such as **Retry** or **Search another Bus**. These improvements provide transparency and help users make informed travel decisions.

6.4 Improve Empty Search Results

Displaying only **"0 Results Found"** does not help users understand why their search failed or what they should do next. Instead, the application should provide a meaningful explanation, such as informing users that the service may have ended, the route is unavailable, or the entered bus number is incorrect. The app should also guide users by offering alternative actions, such as searching nearby buses, viewing the next available service, checking tomorrow's schedule, or modifying the search. This prevents users from reaching a dead end and improves their overall experience.

6.5 Add Contextual Help

Many users may not understand terms such as **Route Number**, **Live Tracking**, or **ETA**, especially first-time users or those unfamiliar with the application. Instead of requiring users to visit a separate help page, the application should provide small contextual help icons beside important fields. These icons can display short explanations and examples whenever users need assistance. Providing guidance exactly where users need it reduces confusion and enables them to complete their tasks more confidently without leaving the current screen.

6.6 Improve Information Hierarchy

The current interface presents multiple pieces of information with equal visual importance, making it difficult for users to identify the most critical details quickly. The application should organize information based on priority. Key information such as the estimated arrival time and bus status should be displayed more prominently using larger fonts or bold styling, while supporting information such as distance, destination, and last updated time should be presented with less emphasis. A clear visual hierarchy allows users to scan the screen quickly and find the information they need with minimal effort.

6.7 Improve Visual Design

The application can become more intuitive by using consistent colours and icons to represent different bus statuses. For example, green can indicate a bus is running, orange can represent delays, red can indicate that the service has ended, and grey can show unavailable information. Similarly, familiar icons for buses, locations, live tracking, and schedules help users recognise information faster without reading large amounts of text. These visual improvements make the interface cleaner, more accessible, and easier to understand.

6.8 Streamline the Overall User Flow

The current user journey contains unnecessary confusion because users are first asked to search using an unclear option and may receive unhelpful search results. A more efficient flow would begin with clear search options such as **Search by Route Number**, **Search by Source and Destination**, or **Search by Bus Number**. The application should then provide auto-suggestions, display live tracking information with accurate updates, and recommend alternative buses if no direct results are available. This streamlined flow reduces user effort, prevents confusion, and helps passengers complete their task more quickly and successfully.

7. Conclusion

By implementing these improvements, the TGSRTC Gamyam application can significantly enhance its usability and user experience. The proposed changes improve clarity, strengthen the visual hierarchy, provide better guidance during errors, and simplify the overall search and tracking process. These enhancements not only reduce user frustration but also increase confidence in the application, enabling passengers to find and track buses more efficiently. The recommendations align with both **Nielsen's Usability Heuristics** and the applicable **UX Laws**, resulting in a more intuitive, accessible, and user-friendly public transport application.